

MAARS Command

Autonomous AI Enterprise Operating System

by MAARS Global Corporation

MAARS Command deploys a workforce of 41 specialized AI agents organized across 8 organizational layers. It merges the power of preset specialist business agents with autonomous execution, persistent memory, and real-world action capabilities. Users set high-level business goals, and the AI workforce autonomously plans, delegates, executes, collaborates, and delivers -- with quality control, failure recovery, and human oversight at every step.

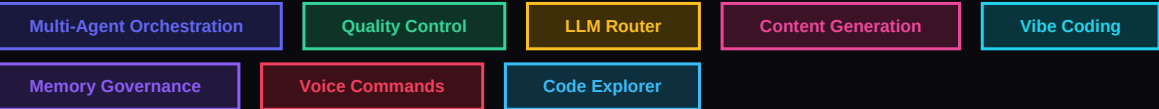


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1. Platform Statistics

AI Agents	41
Organizational Layers	8
Core Systems	17
AI Providers	9
AI Models Available	30+
API Endpoints	212+
Frontend Pages	20+
Database Collections	27+
Total Projects	4
Total Tasks Executed	94
Chat Conversations	140
Memory Entries	4 (3 auto-learned)

2. The 41-Agent AI Workforce

Each agent has a unique identity and Custom Brain Profile: LLM model, autonomy level (1-10), communication style, tool permissions, and creativity temperature. Organized into 8 enterprise layers.

Executive Layer (5 agents)

Commander Orion	[Commander]	Default: gpt-5.2
Supreme orchestrator. Decomposes goals into 3-7 tasks, assigns to specialists, coordinates multi-agent execution, compiles deliverables. <i>Capabilities: Task Delegation, Strategic Planning, Team Orchestration, Goal Breakdown</i>		
Chief Strategy Officer	[Strategist]	Default: gpt-5.2
Markets, competitors, SWOT, business plans, growth roadmaps. Data-backed strategic recommendations. <i>Capabilities: Business Strategy, Competitive Analysis, Market Research</i>		
Revenue Strategist	[Revenue]	Default: gpt-5.2
Monetization modeling, pricing tiers, subscription models, upsell strategies, revenue forecasts. <i>Capabilities: Revenue Optimization, Pricing Strategy, Financial Forecasting</i>		
Investor Relations	[IR]	Default: gpt-5.2
Pitch decks, fundraising materials, quarterly updates, investor comms, financial storytelling. <i>Capabilities: Investor Communications, Fundraising, Pitch Decks</i>		
Business Strategist	[Strategy]	Default: gpt-5.2
Market positioning, business model innovation, go-to-market strategy, market expansion. <i>Capabilities: Market Positioning, Business Models, GTM Strategy</i>		

Product & Technical Layer (7 agents)

Product Manager	[PM]	Default: gpt-5.2
Roadmaps, user stories, RICE/MoSCoW prioritization, sprint planning, product lifecycle. <i>Capabilities: Product Roadmaps, Feature Prioritization, Sprint Planning</i>		
Project Manager	[Project]	Default: gpt-5.2
Timelines, Gantt charts, dependencies, blockers, resource management, milestone tracking. <i>Capabilities: Timeline Management, Risk Management, Resource Allocation</i>		
App Developer	[Dev]	Default: gpt-5.2
Full-stack: React, Node.js, Python, TypeScript. Production-ready web/mobile/API code. <i>Capabilities: Web Development, API Design, Full-Stack, React/Node.js</i>		
Automation Engineer	[Auto]	Default: gpt-5.2
CI/CD pipelines, workflow automation, process optimization, integration scripts. <i>Capabilities: Workflow Automation, CI/CD, Process Optimization</i>		
AI Optimizer	[AI]	Default: gpt-4o
ML model tuning, prompt engineering, cost optimization, model selection. <i>Capabilities: ML Tuning, Performance, Cost Monitoring, Prompt Engineering</i>		
Data Engineer	[Data]	Default: gpt-5.2
Data pipelines, ETL, database architecture, data warehousing, clean data flows. <i>Capabilities: Data Pipelines, ETL, Database Architecture</i>		
Cybersecurity	[Security]	Default: gpt-5.2
Security audits, vulnerability assessments, penetration testing, compliance. <i>Capabilities: Security Audits, Vulnerability Assessment, Compliance</i>		

Creative & Brand Layer (7 agents)

Brand Architect	[Brand]	Default: gpt-5.2
Brand identity systems, positioning, visual language, brand books, style guides. <i>Capabilities: Brand Identity, Positioning, Visual Language</i>		
Graphics Designer	[Design]	Default: gpt-5.2
Images, visual assets, infographics, social graphics, color theory, composition. <i>Capabilities: Image Creation, Visual Assets, Infographics</i>		
Video Specialist	[Video]	Default: gpt-5.2
Scripts, storyboards, YouTube/TikTok formats, motion graphics. <i>Capabilities: Video Scripts, Storyboarding, Motion Graphics</i>		
Copywriter	[Copy]	Default: gpt-5.2
Sales copy, website content, product descriptions, email sequences, AIDA/PAS. <i>Capabilities: Sales Copy, Website Content, Email Sequences</i>		
Web Designer	[Web]	Default: gpt-5.2
Wireframes, landing pages, UI/UX, color palettes, typography, conversion design. <i>Capabilities: UI/UX Design, Wireframing, Landing Pages</i>		
UX Researcher	[UXR]	Default: gpt-5.2
User testing, journey maps, usability analysis, persona development. <i>Capabilities: User Testing, Journey Mapping, Persona Development</i>		
3D Specialist	[3D]	Default: gpt-5.2
3D models, product visualization, AR/VR, immersive content. <i>Capabilities: 3D Modeling, Product Visualization, AR/VR</i>		

Growth & Marketing Layer (7 agents)

Marketing Specialist	[Marketing]	Default: gpt-5.2
Campaigns, ad copy, consumer psychology, viral content, conversion optimization. <i>Capabilities: Campaign Strategy, Performance Marketing, Ad Copy</i>		
Growth Hacker	[Growth]	Default: gpt-5.2
Viral loops, A/B tests, conversion funnels, growth experiments. <i>Capabilities: Viral Loops, A/B Testing, Conversion Optimization</i>		
SEO Specialist	[SEO]	Default: gpt-5.2
Keyword research, on-page/off-page, technical SEO, content strategy. <i>Capabilities: Keyword Research, Technical SEO, Content Strategy</i>		
Social Media Manager	[Social]	Default: gpt-5.2
Platform strategy, community, content calendars, TikTok/IG/LinkedIn algorithms. <i>Capabilities: Platform Strategy, Community Management</i>		
Email Marketing	[Email]	Default: gpt-5.2
Drip campaigns, newsletters, segmentation, deliverability, automated flows. <i>Capabilities: Drip Campaigns, Newsletters, Segmentation</i>		
Sales Representative	[Sales]	Default: gpt-5.2
Lead qualification, outreach sequences, pipeline, B2B/B2C sales. <i>Capabilities: Lead Qualification, Outreach, Pipeline Analysis</i>		
PR Manager	[PR]	Default: gpt-5.2
Media relations, press releases, crisis comms, thought leadership. <i>Capabilities: Media Relations, Press Releases, Crisis Comms</i>		

Operations Layer (6 agents)

Operations Manager	[Ops]	Default: gpt-5.2
Process optimization, SOPs, workflow diagrams, capacity planning. <i>Capabilities: Process Optimization, SOPs, Capacity Planning</i>		
Inventory Manager	[Inventory]	Default: gpt-5.2
Stock levels, supply chain, demand forecasting, warehouse logistics. <i>Capabilities: Stock Management, Supply Chain, Forecasting</i>		
Procurement Manager	[Procurement]	Default: gpt-5.2
Vendor management, cost negotiation, sourcing strategies. <i>Capabilities: Vendor Management, Cost Negotiation, Sourcing</i>		
HR Specialist	[HR]	Default: gpt-5.2
Recruitment, onboarding, culture, performance reviews, HR policy. <i>Capabilities: Recruitment, Onboarding, Performance Reviews</i>		
Customer Service	[Support]	Default: gpt-5.2
Tickets, FAQs, complaints, de-escalation, proactive support. <i>Capabilities: Ticket Resolution, FAQ, Customer Satisfaction</i>		
CX Architect	[CX]	Default: gpt-5.2
Journey optimization, NPS, loyalty programs, touchpoint design. <i>Capabilities: Journey Optimization, NPS, Loyalty Programs</i>		

Finance Layer (2 agents)

Financial Analyst	[Finance]	Default: gpt-5.2
Financial models, DCF valuations, forecasts, budgets, P&L analysis. <i>Capabilities: Financial Modeling, Forecasting, Valuation</i>		
Data Analyst	[Analytics]	Default: gpt-5.2
Dashboards, trend analysis, data visualization, business intelligence. <i>Capabilities: Business Intelligence, Dashboards, Data Viz</i>		

Governance Layer (3 agents)

Legal Assistant	[Legal]	Default: gpt-5.2
Contract review, legal research, compliance, document drafting. <i>Capabilities: Contract Review, Legal Research, Compliance</i>		
Compliance Officer	[Compliance]	Default: gpt-5.2
Regulatory adherence, audit prep, policy monitoring, frameworks. <i>Capabilities: Regulatory Adherence, Audit Preparation</i>		
Ethics Officer	[Ethics]	Default: gpt-5.2
Ethical AI governance, bias detection, fair practices. <i>Capabilities: Ethical AI, Bias Detection, Responsible AI</i>		

Intelligence Layer (4 agents)

Research Specialist	[Research]	Default: gpt-5.2
Deep research, competitive analysis, trend forecasting, market intelligence. <i>Capabilities: Deep Research, Competitive Analysis, Trends</i>		
Knowledge Architect	[Knowledge]	Default: gpt-5.2
Knowledge bases, taxonomy design, information architecture. <i>Capabilities: Knowledge Base, Taxonomy, Info Architecture</i>		
Localization Specialist	[L10n]	Default: gpt-5.2
Translation, cultural adaptation, multi-language content, 50+ languages. <i>Capabilities: Translation, Cultural Adaptation, Multi-Language</i>		

Personal Secretary [\[Assistant\]](#) Default: gpt-5.2

Calendars, emails, scheduling, real-world action bridge from Commander.
Capabilities: Calendar, Email, Scheduling, Action Execution

3. Core Systems & Capabilities (17 Systems)

These are the engines powering MAARS Command. Each system has its own data model, API endpoints, and configuration.

1. Autonomous Orchestration Engine

Commander Orion receives a business goal, scores complexity (1-10), decomposes into 3-7 milestones with tasks, assigns to optimal specialists, executes in parallel where dependencies allow, applies quality control, and compiles a unified deliverable. Supports failure recovery and escalation.

Endpoints: POST /api/projects, POST /api/projects/{id}/execute, GET /api/projects/{id}

2. Custom Brain Profiles

Every agent has a configurable brain: LLM model selection (9 providers), autonomy level (1-10), communication style (formal/casual/technical/creative), creativity temperature (0.0-1.0), tool permissions (web search, code execution, email), and context window size. Profiles persist per-user.

Endpoints: GET/PUT/DELETE /api/agents/{id}/brain, GET /api/brain-profiles

3. Quality Control & Failure Recovery

Critic Module (GPT-4o) auto-reviews every output on a 1-10 scale (accuracy, completeness, relevance). Failed tasks auto-retry with fallback chain: GPT-5.2 -> Claude Sonnet -> Gemini Pro -> GPT-4o. If all fail, escalates to Commander then user.

Endpoints: POST /api/enterprise/quality/review, GET /api/enterprise/quality/history

4. Model-Agnostic LLM Router

Analyzes task content for complexity signals (coding, reasoning, creative, data, legal, simple). Maps to tiers: Premium (GPT-5.2, Claude), Standard (GPT-4o, Gemini Flash), Economy (GPT-4o Mini, Haiku). Logs all decisions. User preference can override.

Endpoints: POST /api/enterprise/router/route-task, GET /api/enterprise/router/history

5. Autonomous Collaboration Engine

After each task, maps agent to 1 of 9 domains. Scans result for cross-domain keywords. Auto-creates collaboration requests to relevant agents. 41 agents across 9 domains with trigger rules for seamless coordination.

Endpoints: POST/GET/PATCH /api/collaborations, GET /api/enterprise/collaboration/log

6. Universal Reference Intelligence

Analyzes reference text to extract Style Blueprints: writing style, tone, vocabulary, sentence patterns, audience profile, content structure, key phrases. Blueprints stored and reusable across Content Generator.

Endpoints: POST /api/reference/analyze, GET /api/reference/blueprints

7. Content Generator

Generates on-brand content using Style Blueprints. 8 types: blog, social, email, ad copy, product desc, press release, newsletter, script. Length control (short/medium/long), tone override, generation history.

Endpoints: POST /api/content/generate, GET /api/content/history

8. Vibe Coding App Builder

Chat-based full-stack app generation. Each message: intent analysis -> code generation (HTML/CSS/JS/Tailwind) -> live iframe preview -> iterative refinement -> download. Persistent projects.

Endpoints: POST /api/vibe/projects, POST /api/vibe/projects/{id}/chat, GET .../preview

9. Agent Activity Monitor (WebSocket)

Real-time dashboard via WebSocket (/api/ws/activity, 8s updates). Shows: agent status with completion rates, inter-agent communication flows, task dependency graph, recent tool executions. REST fallback polling at 10s.

Endpoints: WS /api/ws/activity, GET /api/enterprise/activity/live

10. Simulation vs. Execution Mode

System-wide toggle. Simulation (default): agents return descriptive responses, no external APIs called. Execution: real Gmail sends, Calendar events, API calls. Confirmation prompt before mode switch.

Endpoints: GET/PUT /api/enterprise/system/mode

11. Real-World Action Layer

Google Suite integration (Gmail + Calendar) via OAuth 2.0. All actions gated by system mode. Personal Secretary as execution bridge. Full action logging.

Endpoints: GET /api/oauth/gmail/login, POST /api/actions/send-email, POST /api/actions/create-event

12. Flexible LLM Configuration

9 providers, 30+ models. Users choose preferred model in Settings. Override priority: Agent brain profile > User default > LLM Router auto-selection.

Endpoints: PUT /api/users/me/config

13. Voice Command Interface

Mic button in Command Palette captures audio (WebM via MediaRecorder), transcribes via OpenAI Whisper, auto-fills search for hands-free navigation across pages, agents, and actions.

Endpoints: POST /api/voice/transcribe

14. Admin Code Explorer

Full codebase browser: recursive file tree, code viewer with line numbers and language detection, file search, copy-to-clipboard, ZIP download of entire codebase. Admin-only with path traversal protection.

Endpoints: GET /api/admin/code/tree, /file, /search, /export

15. Memory Governance

CRUD with versioning (every update creates version). Relevance scoring: time-decay (30-day half-life) + access frequency + importance weight. Auto-pruning with dry-run preview. 500-entry limit. Stats dashboard.

Endpoints: POST/GET/PUT/DELETE /api/memory/entries, POST /api/memory/prune, GET /api/memory/stats

16. Agent Memory Auto-Learning

GPT-4o-mini extracts 1-3 key learnings from every completed task. Categories: fact, preference, instruction, context, decision. Near-duplicate detection. Async execution (never blocks task completion).

Endpoints: Auto-triggered by orchestration_service.py and agent_service.py

17. Command Palette

VS Code-style quick search (press / or Cmd+K). Searches 20+ pages, 41 agents, and quick actions. Keyboard navigation, recent searches in localStorage, voice integration via mic button.

Endpoints: Frontend-only (uses /api/agents/public for agent data)

4. AI Models & Providers (9 Providers)

9 AI providers with 30+ models for text, reasoning, search, image generation, video, voice, and speech-to-text. All accessible via Emergent Universal LLM Key or direct API keys.

OpenAI (8 models)

Model	Tier	Cost (In/Out per 1M tok)
GPT-5.2	Flagship	\$2.50/\$10
GPT-4o	Fast	\$2.50/\$10
GPT-4o Mini	Economy	\$0.15/\$0.60
O3	Reasoning	\$10/\$40
O3 Mini	Reasoning	\$1.10/\$4.40
GPT Image 1	Image	\$0.02/img
Sora 2	Video	\$0.10/sec
Whisper	STT	Included

Anthropic (3 models)

Model	Tier	Cost (In/Out per 1M tok)
Claude Sonnet 4.5	Flagship	\$3/\$15
Claude Opus 4.5	Premium	\$15/\$75
Claude Haiku 4.5	Economy	\$0.80/\$4

Google Gemini (3 models)

Model	Tier	Cost (In/Out per 1M tok)
Gemini 3 Flash	Fast	\$0.075/\$0.30
Gemini 3 Pro	Flagship	\$1.25/\$5
Nano Banana 2	Image	\$0.02/img

xAI (Grok) (3 models)

Model	Tier	Cost (In/Out per 1M tok)
Grok 3	Flagship	\$3/\$15
Grok 3 Mini	Economy	\$0.30/\$0.50
Grok 2	Fast	\$2/\$10

DeepSeek (2 models)

Model	Tier	Cost (In/Out per 1M tok)
DeepSeek Chat	Economy	\$0.14/\$0.28
DeepSeek Reasoner	Reasoning	\$0.55/\$2.19

Mistral AI (3 models)

Model	Tier	Cost (In/Out per 1M tok)
Mistral Large	Flagship	\$2/\$6
Mistral Medium	Fast	\$0.40/\$2
Mistral Small	Economy	\$0.10/\$0.30

Perplexity (2 models)

Model	Tier	Cost (In/Out per 1M tok)
Sonar	Search	\$1/\$1
Sonar Pro	Research	\$3/\$15

Cohere (2 models)

Model	Tier	Cost (In/Out per 1M tok)
Command R+	Flagship	\$2.50/\$10
Command R	Economy	\$0.15/\$0.60

ElevenLabs (2 models)

Model	Tier	Cost (In/Out per 1M tok)
Multilingual v2	Voice	\$0.30/1K ch
Turbo v2.5	Voice	\$0.18/1K ch

5. Technical Architecture

Backend Stack

- FastAPI (Python 3.11) -- async/await, high-performance ASGI
- MongoDB -- 27+ collections (agents, tasks, projects, chats, memories...)
- WebSocket -- real-time Activity Monitor streaming (8s updates)
- JWT authentication with admin role detection
- Emergent Integrations SDK -- unified 9-provider LLM access
- Background tasks via `asyncio.create_task` (non-blocking)
- StreamingResponse for ZIP/PDF file downloads

Frontend Stack

- React 18 with hooks and Context API
- Tailwind CSS with custom dark theme
- Shadcn/UI component library (Button, Card, Badge, etc.)
- React Router v6 with protected and admin routes
- DashboardLayout -- single resizable sidebar for all 20+ pages
- Command Palette with voice integration
- WebSocket client with REST fallback

AI & LLM Layer

- 9 providers: OpenAI, Anthropic, Google, xAI, DeepSeek, Mistral, Perplexity, Cohere, ElevenLabs
- 30+ models: text, reasoning, search, image, video, voice, STT
- Model-Agnostic LLM Router (auto-selects optimal model)
- Quality Control Critic Module (GPT-4o, 1-10 scoring)
- Failure Recovery (GPT-5.2 -> Claude -> Gemini -> GPT-4o)
- Memory Auto-Learning (GPT-4o-mini knowledge extraction)
- Universal Emergent LLM Key (single key, all providers)

Database (27+ Collections)

- users, agents, agent_brains -- Accounts, definitions, brain profiles
- chats, messages -- Conversation history with agents
- tasks, projects -- Task execution and project orchestration
- collaborations -- Inter-agent collaboration requests
- quality_reviews -- Quality control scores and feedback
- memory_entries -- Versioned memory with relevance scoring
- vibe_projects -- Vibe Coding app builder projects
- style_blueprints, generated_content -- Reference Intelligence
- teams, team_invites -- Team management
- subscriptions, usage_logs -- Billing and usage tracking
- routing_logs -- LLM Router decision audit trail
- oauth_tokens, action_logs -- Google OAuth and action logs
- system_config -- System-wide settings

6. Complete API Reference (212+ Endpoints)

All endpoints require JWT auth (Authorization: Bearer <token>) unless noted. Admin endpoints require is_admin=true.

Authentication

POST	/api/auth/register
POST	/api/auth/login
GET	/api/auth/me
GET	/api/stats

Agents (41)

GET	/api/agents/public
GET/POST	/api/agents
GET	/api/agents/{id}
POST	/api/agents/{id}/chat
GET/PUT/DELETE	/api/agents/{id}/brain

Projects & Orchestration

POST	/api/projects
GET	/api/projects
POST	/api/projects/{id}/execute
GET	/api/projects/{id}
PATCH/DELETE	/api/projects/{id}

Tasks

GET	/api/tasks
POST	/api/tasks
PATCH	/api/tasks/{id}
POST	/api/tasks/{id}/execute

Enterprise

POST/GET/PATCH	/api/collaborations
GET	/api/enterprise/kpis
GET	/api/enterprise/activity/live
WS	/api/ws/activity
GET/PUT	/api/enterprise/system/mode
POST	/api/enterprise/quality/review
POST	/api/enterprise/router/route-task

Memory

POST/GET	/api/memory/entries
PUT/DELETE	/api/memory/entries/{id}
GET	/api/memory/entries/{id}/versions
POST	/api/memory/prune
GET	/api/memory/stats

Content & Intelligence

POST	/api/reference/analyze
GET	/api/reference/blueprints
POST	/api/content/generate
GET	/api/content/history

Vibe Coding

POST	/api/vibe/projects
GET	/api/vibe/projects
POST	/api/vibe/projects/{id}/chat
GET	/api/vibe/projects/{id}/preview

● Voice & Code Explorer

POST	/api/voice/transcribe
GET	/api/admin/code/tree
GET	/api/admin/code/file
GET	/api/admin/code/search
GET	/api/admin/code/export
GET	/api/summary/pdf

● Real-World Actions

GET	/api/oauth/google/login
GET	/api/oauth/google/callback
POST	/api/actions/send-email
POST	/api/actions/create-event

● Teams

POST	/api/teams
GET	/api/teams
POST	/api/teams/{id}/invite
PUT	/api/teams/{id}/members/{uid}
POST	/api/chats/{id}/share

● Billing

GET	/api/plans
GET	/api/subscription
POST	/api/checkout
GET	/api/credits

● Admin (30+)

GET	/api/admin/stats
GET	/api/admin/users
GET	/api/admin/analytics
GET	/api/admin/profit
GET	/api/admin/audit-log
GET	/api/admin/agent-performance



MAARS Command

End of Documentation

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